

Chain rule



1 $y = (2x + 3)^3$

2

a $u(x) = 5x^3 - 4x^2 + 3x - 5, f(u) = u^7$

b $u(x) = 2x^2 + 2x + 3, f(u) = \sqrt[4]{u}$

c $u(x) = 4x^2 - 3x + 5, f(u) = \frac{1}{u^3}$

3

a $\frac{du}{dx} = 8x^3$

b $y = u^5$

c $\frac{dy}{du} = 5u^4$

d $\frac{dy}{dx} = 40x^3 (2x^4 + 6)^4$

4

a i $\frac{dy}{du} = 5u^4$

ii $\frac{du}{dx} = 1$

iii $\frac{dy}{dx} = 5(x + 5)^4$

b i $\frac{dy}{du} = -u^{-2}$

ii $\frac{du}{dx} = 4$

iii $\frac{dy}{dx} = -4(4x + 3)^{-2}$

c i $\frac{dy}{du} = \frac{1}{2\sqrt{u}}$

ii $\frac{du}{dx} = 2x$

iii $\frac{dy}{dx} = \frac{x}{\sqrt{5 + x^2}}$

d i $\frac{dy}{du} = \frac{1}{3}u^{-\frac{2}{3}}$

ii $\frac{du}{dx} = -15x^{-2}$

iii $\frac{dy}{dx} = -5 \times 15^{-\frac{2}{3}} x^{-\frac{4}{3}}$

5

a $\frac{dy}{dx} = 50x - 70$

b $\frac{dy}{dx} = 10(50x - 7)$

6 $\frac{dy}{dx} = \frac{-2x}{(x^2 + 5)^2}$

7 $\frac{dy}{dx} = \frac{4}{\sqrt{8x + 5}}$



- a $\frac{dy}{dx} = 36(4x + 3)^8$
- b $\frac{dy}{dt} = -4(2t^7 + 8t^3 + 3t + 5)^{-5}(14t^6 + 24t^2 + 3)$
- c $\frac{dy}{dx} = 3(x^2 + x^{-3})^2(2x - 3x^{-4})$
- d $\frac{dy}{dx} = \frac{1}{3}(x^2 - 5x)^{-\frac{2}{3}}(2x - 5)$
- e $\frac{dy}{dx} = 8(3x^2 - 4x + 2)^3(3x - 2)$
- f $\frac{dy}{dx} = -90(3x + 4)^9$
- g $\frac{dy}{dx} = 4(9x + 7)^{\frac{1}{3}}$
- h $\frac{dy}{dx} = -5(x + 6)^{-6}$
- i $\frac{dy}{dx} = -(3x^3 - 12x^2 + 5)(x^4 - 4x^3 + 5x)^{-2}$
- j $\frac{dy}{dx} = -(1 + x)^{-\frac{3}{2}}$
- k $\frac{dy}{dx} = \frac{24}{5}(4x + 1)^{\frac{1}{5}}$
- l $\frac{dy}{dx} = 8\left(\frac{1}{3}x + 1\right)^{-7}$
- m $\frac{dy}{dx} = -\frac{5}{6}\left(4 - \frac{1}{3}x\right)^{-\frac{1}{2}}$
- n $\frac{dy}{dx} = 2\sqrt{5}(1 - x\sqrt{5})^{-2}$
- o $\frac{dy}{dx} = 4\left(x^{-\frac{1}{2}} - x^{-\frac{3}{2}}\right)\left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right)^7$
- p $\frac{dy}{dx} = \frac{4}{3}(9 - x)^{-\frac{7}{3}}$
- q $\frac{dy}{dx} = -15(1 - x^{-2})(x + x^{-1})^4$



Gradients, tangents and normals

9 $g'(-1) = -256$

10

a $-\frac{3}{2}$

b Increasing

c $-\frac{\sqrt{21}}{3} < x < 0$

11

a $\frac{dy}{dx} = -(3 - 2x)^{-\frac{1}{2}}$

b $\frac{dy}{dx} = 0$

c **C**

d The gradient of the tangent gets close to horizontal but never reaches it. There is no value of x where $\frac{1}{\sqrt{3-2x}} = 0$.

12 $f'(8) = 27$

13 36

14 $f'(2) = 3240$

15 $x = 5$

16 $x = 2, x = -6$

17 $x = -1, 0, 1$

18 $x = -2, -1, 0$

19 $(4, 1)$

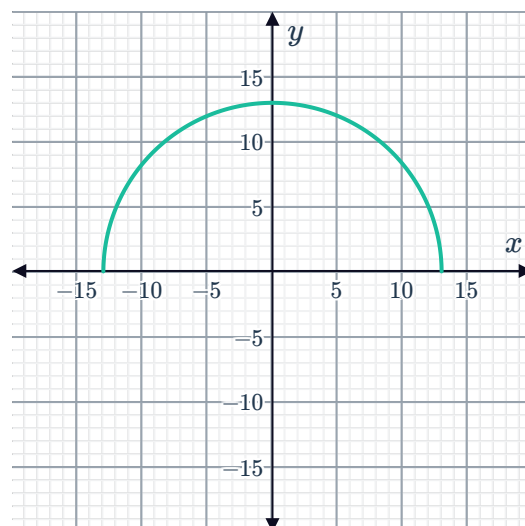
20 $x = 1, 6$

21

a $\frac{dy}{dx} = -x(169 - x^2)^{-\frac{1}{2}}$



b



c $5x + 12y = 169$

22 $y = -8x - 7$

23 $y = 16x - 15$

24 $y = -\frac{x}{9} - \frac{8}{9}$

25 $y = -\frac{x}{64} + \frac{1025}{64}$

26

a $y = -2x - 1$

b $y = \frac{x}{2} + \frac{3}{2}$